# Topic 2: Introduction to Forecasting

*ECON 4753 — University of Arkansas*

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# Outline

**Forecasting**

Goals of Forecasting

Evaluating Models

Types of Data

# Problem of Prediction

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- Learn about who are potential customers to advertise to based on their observable characteristics
  - → Input: observable characteristics
  - → Outcome: whether they purchase a product
- Predict values of a variable in the future, e.g. time-series of stock prices
  - → Input: the time-period
  - → Output: stock price

# Prediction model

We have an outcome variable $Y$ and a set of $p$ different predictor variables $X = (X_1, X_2, \ldots, X_p)$.

- For some observations we observe both $X$ and $Y$; this is essential to fit the model

We can write the true prediction model in a general form as

$$Y = f(X) + \varepsilon,$$

where $f$ is some unknown function of $X$ and $\varepsilon$ is the error term which we assume is unrelated to $X$ and mean zero.

# Limits of prediction

$$Y = f(X) + \varepsilon,$$

You should think of the function $f$ as being an 'oracle' who is as good of a forecaster as possible given the information $X$. Even then, there remains $\varepsilon$ which is unpredictable given $X$

- E.g. predicting wages given gender and education levels. $f$ gives us our best guess at wages, but there remains much variation in wages that are not predictable.

# Modelling strategies

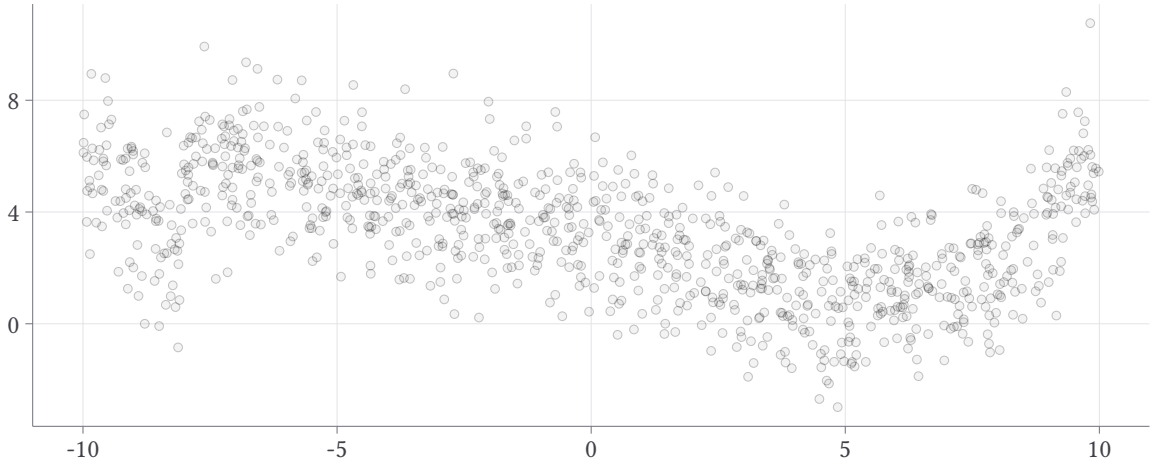
The goal of the class will be to teach you how to do good at modelling the function $f$

- How do we best try to learn the true systematic relationship between $X$ and $y$

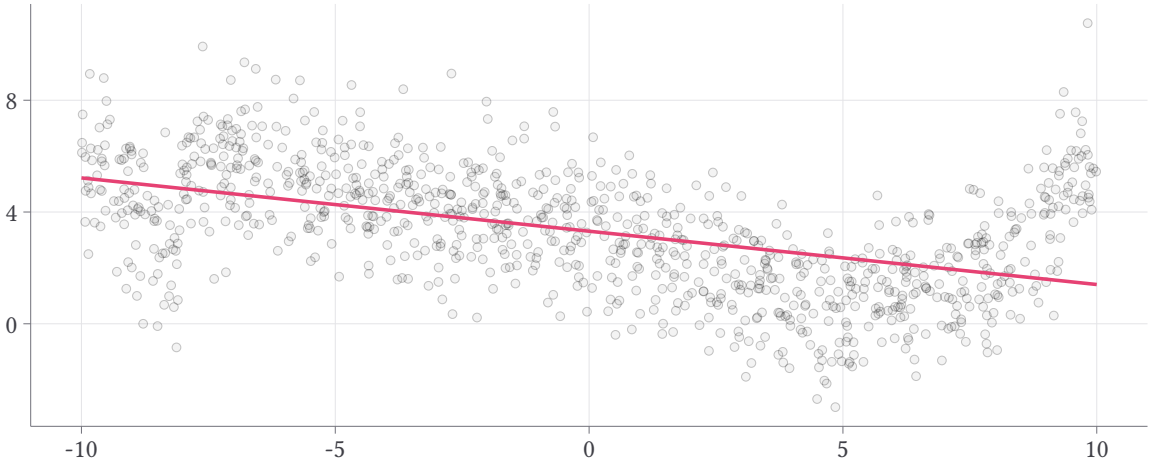
Two ways to get better at modelling (in order of importance):

1. Become a subject matter expert so that you have great intuition on the relationship between $y$ and $X$
2. (this class) Fill your toolbox with a bunch of different modelling strategies that you know when to reach for

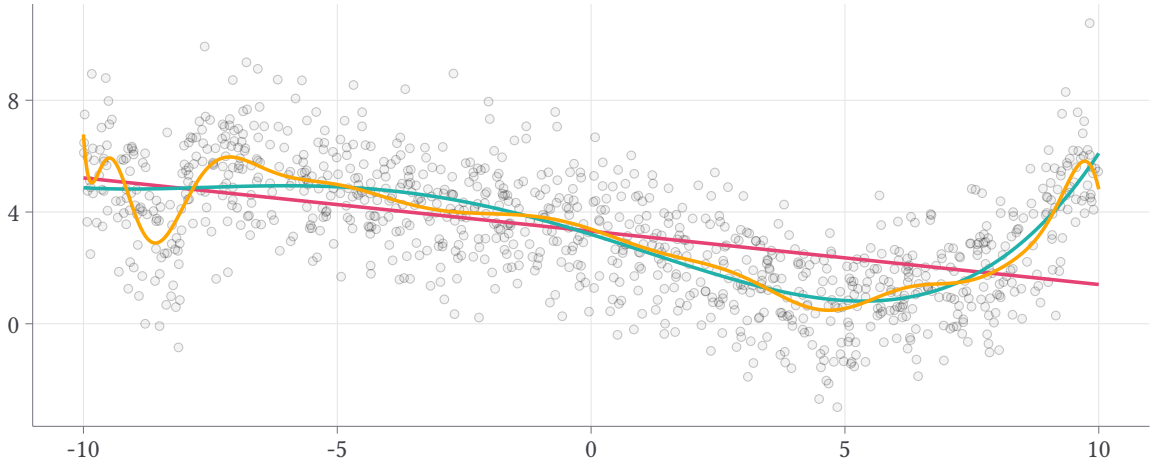
Models for $f$:



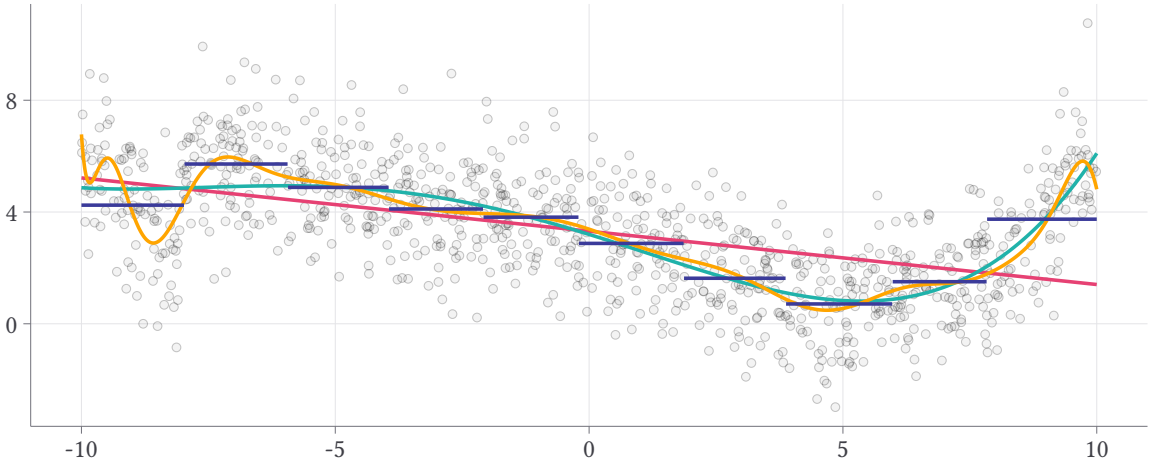
Models for $f$: Line

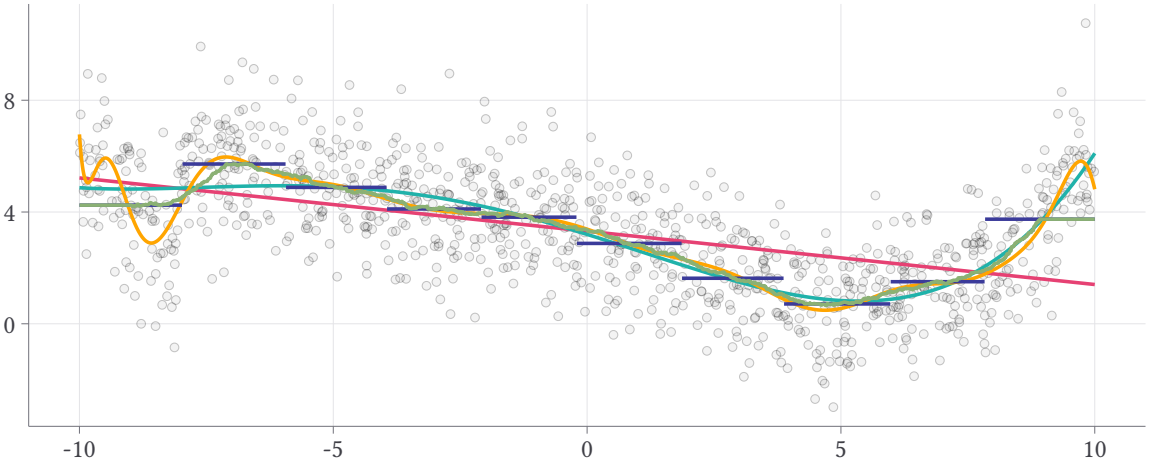


Models for $f$: Line, Polynomial $(x^4)$



Models for $f$: Line, Polynomial $(x^4)$, Bins of $x$




Models for $f$: Line, Polynomial $(x^4)$, Bins of $x$, KNN of $x$
8
4
0
-10
-5
0
5
10

# Models with multiple variables: Advertising Example

The previous slides showed examples of how to think about modelling the relationship between a single variable and the outcome

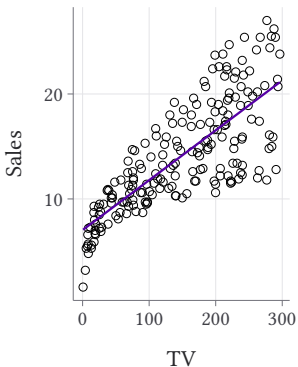
Things get more complicated when we have many different variables we can use to predict $y$

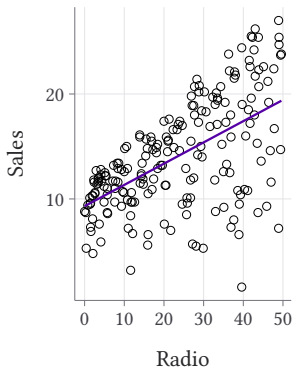
# Advertisting Example

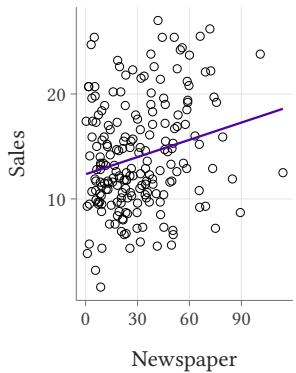
Let’s given an example. Say you’re a business and you want to use advertising to boost sales. You have a bunch of different markets (e.g. cities) and you have data on how you’ve spent your advertising budget in those markets and the sales in that market

# Advertising Example

*Single-variable predictors*



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We see that sales are higher in markets that have spending on TV, Radio, and Newspaper ads separately.

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These single scatter plots with line of best-fits are a somewhat poor model:

- Are there synergies between different advertising strategies (are they substitutes or complements to one another)?
- Do places with more TV ads also have more radio ads? Then how can we tell if it is TV ads that are helping or if it is really radio ads

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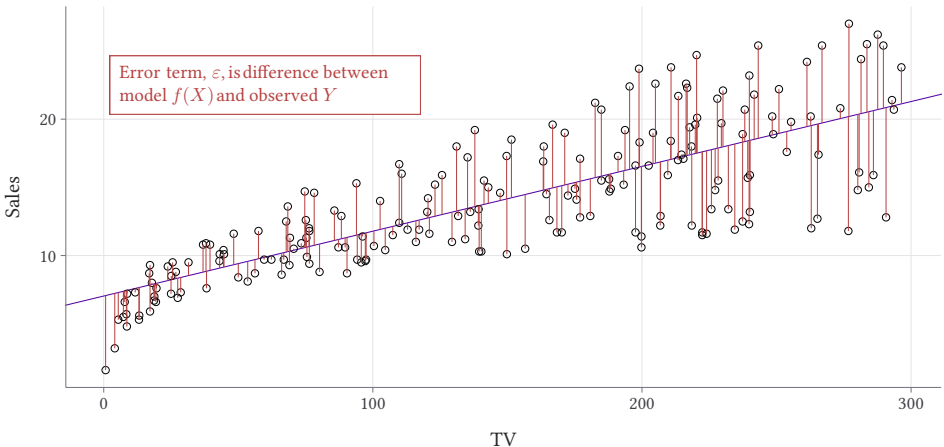
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Key takeaway: Forecasting models get better the more carefully you think about the context you are in

# Error term


Error term, $\varepsilon$, is difference between model $f(X)$ and observed $Y$
Sales
20
10
0
100
200
300
TV

# Error term

In the previous figure, we were able to determine $\varepsilon$ because we assumed we *knew* the model $f$ and therefore could observe $f(X)$ for each market.

In reality, we do not know $f$ and can never observe $\varepsilon$. But, we can try and estimate it ...

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**Goals of Forecasting**

Evaluating Models

Types of Data

# Why estimate $f$?

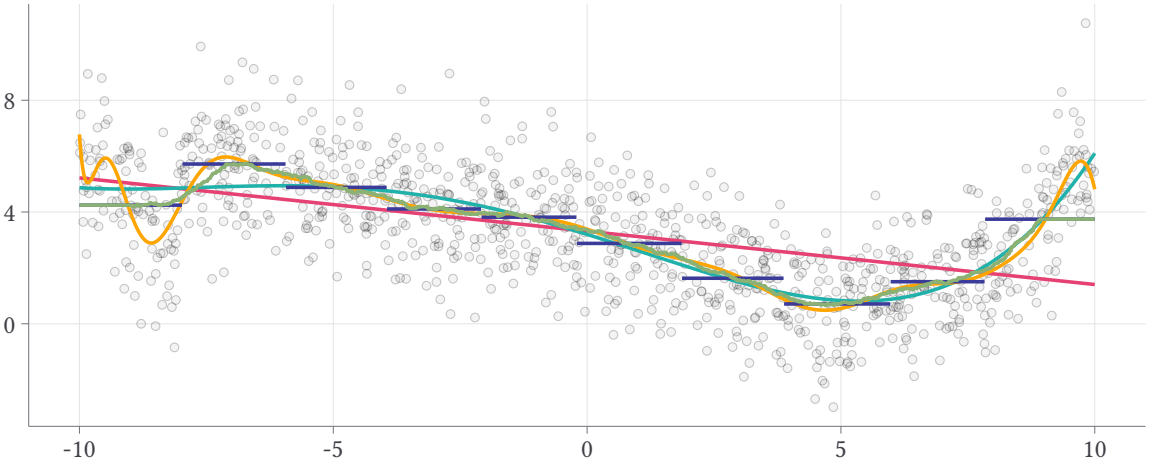
There are two related reasons to try and predict $f$:

1. Predict $y$ as good as possible (prediction)
   - $\rightarrow$ Think of prediction as a 'black box' where the goal is to do as good of a job at predicting $y$ as possible
2. Understand the relationship between $x$ and $y$ (inference)
   - $\rightarrow$ If our goal is being able to describe the relationship between $x$ and $y$.

# Model Flexibility

There is a limit to how flexible we can make our model

1. If our goal is prediciton, we only have a finite amount of data to use to fit the model, so there's a limit on how much we can learn
   $\rightarrow$ Face the risk of overfitting the data (chasing after the random noise $\varepsilon$)
2. If our goal is inference, then added flexibility is harder to summarize to stakeholders.


Models for $f$: Line, Polynomial $(x^4)$, Bins of $x$, KNN of $x$
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# Prediction Error

Given our model, we will want to be able to evaluate how good our model does at predicting observations $y$

Define the prediction error as

$$\hat{\varepsilon} = \underbrace{y}_{\text{true value}} - \underbrace{\hat{y}}_{\text{predicted value}}$$

# Prediction Error

$$\hat{\varepsilon} = \underbrace{y}_{\text{true value}} - \underbrace{\hat{y}}_{\text{predicted value}}$$

Large $\hat{\varepsilon}$ means you did a poor job of predicting that observation. That could be because

1. The linear model is bad at predicting $y$
2. Or, the true noise $\varepsilon$ is making $y$ far away from the systematic component $f(X)$ for this observation

# Mean-square prediction error

To provide a summary measure of fit, we want to *average* prediction error over many observations. This will find a ‘average’ prediction error

- If we took the simple mean of prediction error, positive and negative prediction errors would cancel out

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The mean-square (prediction) error (MSE) is calculated as:

$$\text{MSE} \equiv \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1}{n}\sum_{i=1}^{n}\hat{\varepsilon}_i^2 \tag{1}$$

- Average of squared prediction error

# Mean-square prediction error

| $y_i$ | $\hat{y}_i$ | $\hat{\varepsilon}_i$ |
|---|---|---|
| 3.7 | 4.20 | |
| 4.1 | 4.18 | |
| 5.6 | 5.48 | |
| 2.9 | 3.29 | |
| 8.8 | 8.81 | |

Calculate mean-square prediction error:

# Mean-square prediction error

| $y_i$ | $\hat{y}_i$ | $\hat{\varepsilon}_i$ |
|---|---|---|
| 3.7 | 4.20 | 0.5 |
| 4.1 | 4.18 | 0.08 |
| 5.6 | 5.48 | -0.12 |
| 2.9 | 3.29 | 0.39 |
| 8.8 | 8.81 | 0.01 |

Calculate mean-square prediction error:

$$\begin{aligned}\text{MSPE} &= \frac{1}{5}\left(0.5^2 + 0.08^2 + -0.12^2 + 0.39^2 + 0.01^2\right) \\ &= 0.0846\end{aligned}$$

# In-sample vs. Out-of-sample prediction error

As a forecaster, you will fit a model using a set of observations $\{(x_1, y_1), \ldots, (x_n, y_n)\}$. This is called the training data.

We can calculate the in-sample MSE by formula (1) averaging over all observations in the training data.

- This tells us how good we do at predicting the data *we trained the model on*.

# In-sample vs. Out-of-sample prediction error

If our goal is prediction, we really want to know how the model would predict on *new* observations that we *have not seen before*

- It is common to hold out a set of test data that is NOT used for training the model, but just for evaluating it's performance

# Why use ‘test data’?

It is common to try and ‘pick’ from a set of models based on how they do at in-sample prediction:

- That is, select the model with the smallest *in-sample MSE*.

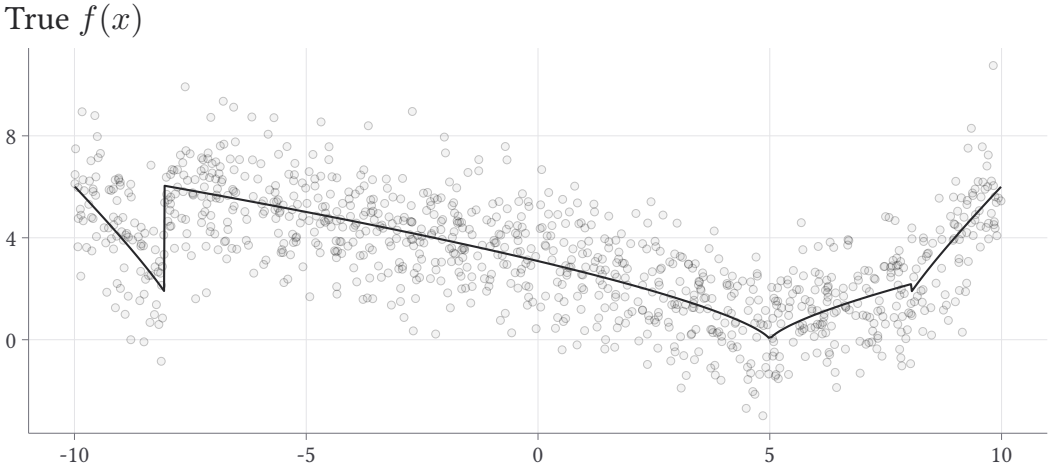
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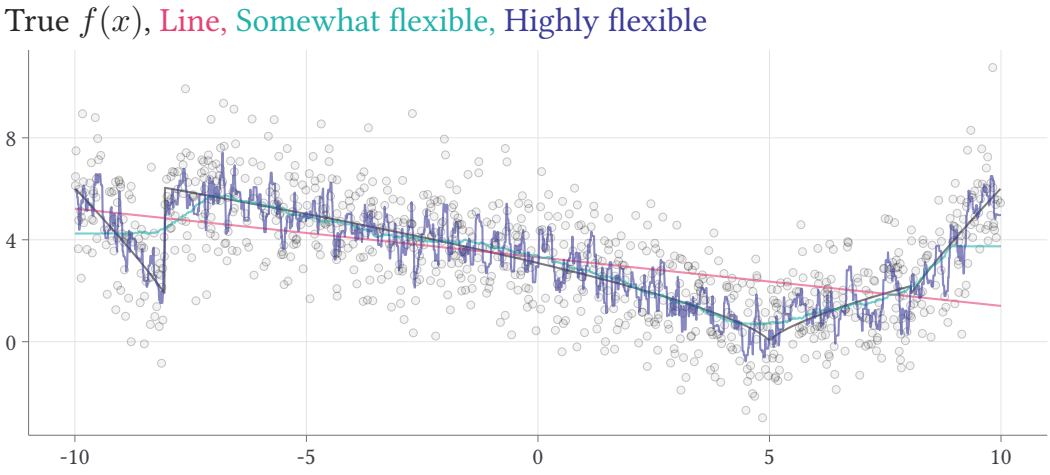
- That is, select the model with the smallest *in-sample MSE*.

This is *a bad thing to do*; by focusing on fitting the current sample very well, you are risking overfitting the data

# Flexibility vs. Overfitting



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By making the model more and more *flexible*, you risk overfitting more and more

- A solution is to evaluate your model fit using outside ‘testing data’

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This technique is not as common when you care more about the associations between variables (interpreting the model)

- Not really a good reason other than "that is more complicated"

# Bias-variance trade-off

This discussing of increasing flexiblity leading to increasing the noise of the model fit is a well-known problem. It is called the Bias-Variance Tradeoff:

1. Bias: When the model we fit, $\hat{f}(x)$, does a poor job fitting the true model $f(x)$
2. Variance: When the model we fit, $\hat{f}(x)$, is very variable across samples
   $\rightarrow$ In repeated sampling, the model we estimate varies from estimate to estimate

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This is a 'trade-off'. To lower bias by adding flexibility, you're adding variance (noisiness) to the estimate

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**Types of Data**

# Cross-sectional Data

Cross-sectional data consists of many different *units* viewed at a point in time:

| school_id | avg_sat_math | pct_white | pct_black |
|---|---|---|---|
| 01M539 | 657 | 28.6% | 13.3% |
| 02M294 | 395 | 11.7% | 38.5% |
| 02M308 | 418 | 3.1% | 28.2% |
| 02M545 | 613 | 1.7% | 3.1% |
| 01M292 | 410 | 3.9% | 24.4% |
| 01M696 | 634 | 45.3% | 17.2% |

# Time-series Data

Time-series data consists of a single unit viewed over multiple points in time:

| month | day | hour | bikers | temp |
|---|---|---|---|---|
| 1 | 1 | 0 | 16 | 0.24 |
| 1 | 1 | 1 | 40 | 0.22 |
| 1 | 1 | 2 | 32 | 0.22 |
| ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |
| 12 | 31 | 21 | 52 | 0.40 |
| 12 | 31 | 22 | 38 | 0.38 |
| 12 | 31 | 23 | 31 | 0.36 |

# Panel Data

Panel data consists of many units observed over multiple points in time:

| hedge_fund_manager | month | return |
|---|---|---|
| 1 | 1 | -3.34% |
| 1 | 2 | 3.76% |
| 1 | 3 | 12.97% |
| ⋮ | ⋮ | ⋮ |
| 2000 | 48 | -3.76% |
| 2000 | 49 | 2.25% |
| 2000 | 50 | 6.68% |